Lesson 45: Policy Iteration

AI/ML Mathematics

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1 Motivation and Overview

Lesson 42 introduced value iteration as the natural algorithm arising from the Banach fixed-point theorem: repeatedly apply the Bellman optimality operator $\mathcal{T}^*$ until convergence. While value iteration is conceptually simple and widely used, its convergence rate γ per iteration means that for γ close to 1 the algorithm can require hundreds or thousands of iterations. Consider a logistics MDP with a discount factor $\gamma = 0.999$: the a priori bound from Lesson 42 predicts roughly 10 000 iterations to reach $\varepsilon = 0.01$ accuracy. Can we do fundamentally better?

Policy iteration, introduced by Howard [1960], takes a radically different approach. Rather than iterating on value functions, it iterates on *policies*: evaluate the current policy exactly (by solving a linear system), then improve the policy by acting greedily with respect to the computed values. Because the number of deterministic stationary policies is finite (A^S), this process must terminate – and it terminates at the optimal policy. In practice, policy iteration converges in remarkably few iterations (often far fewer than A^S), with each iteration making a large improvement. The tools developed in this lesson – the policy evaluation step, the policy improvement theorem, Howard's improvement principle, and modified policy iteration – complete the algorithmic toolkit for solving known MDPs.

Why Policy Iteration Matters for RL/ML

- **Actor-critic methods**: The actor-critic architecture directly mirrors policy iteration: the critic performs approximate policy evaluation, and the actor performs (approximate) policy improvement.
- **Generalized policy iteration (GPI)**: Sutton and Barto identify the GPI principle – alternating evaluation and improvement – as the unifying idea behind nearly all RL algorithms.
- **Exact planning**: For moderate-sized MDPs with known models, policy iteration is often the algorithm of choice due to its rapid convergence, independent of γ.
- **Natural policy gradient**: The natural policy gradient can be interpreted as a soft version of policy improvement in a parameterized policy space, connecting PI to modern deep RL.

Prerequisites. MDP model, stationary policies, value functions V^π and Q^π (Lesson 39). Bellman expectation operator $\mathcal{T}^\pi$, Bellman optimality operator $\mathcal{T}^*$, contraction properties, the policy improvement theorem (Lesson 41, Theorem 7.3). Value iteration algorithm, convergence, error bounds (Lesson 42). Linear systems and matrix inversion (Lessons 5–6).

Learning objectives. After completing this lesson, the student will be able to:

- *Derive* the policy evaluation step as the solution of a linear system $v = (I - \gamma P^\pi)^{-1} r^\pi$.
- *State* and *prove* the policy improvement theorem in its general form.
- *Prove* that policy iteration converges to the optimal policy in at most A^S iterations.
- *Explain* Howard's improvement principle: strict improvement at every non-terminal iteration.
- *Describe* modified policy iteration and *prove* its convergence.
- *Compare* the computational trade-offs between value iteration and policy iteration.

2 References

- Puterman [2014, Chapter 6, Sections 6.4–6.5] – Definitive treatment of policy iteration, modified policy iteration, and convergence analysis for discounted MDPs.
- Howard [1960] – Original monograph introducing policy iteration and proving its finite convergence.
- Bertsekas [2012, Chapter 1, Sections 1.3–1.4] – Policy iteration, modified policy iteration, and connections to approximate dynamic programming.
- Sutton and Barto [2018, Chapter 4] – Accessible presentation of policy iteration and the generalized policy iteration principle.
- Denardo [1967] – Unified contraction framework placing both value iteration and policy iteration within a common operator-theoretic setting.

3 Policy Evaluation

We work throughout with a finite MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma)$ with $|\mathcal{S}| = S$, $|\mathcal{A}| = A$, $R_{\max} = \max_{s,a} |R(s,a)|$, and $\gamma \in [0,1)$. For a deterministic stationary policy $d : \mathcal{S} \to \mathcal{A}$, we write $P^d \in \mathbb{R}^{S \times S}$ for the transition matrix $P^d(s,s') = P(s' \mid s, d(s))$ and $r^d \in \mathbb{R}^S$ for the reward vector $r^d(s) = R(s, d(s))$.

Definition 3.1 (Policy Evaluation Problem)**.** Given a deterministic stationary policy d, the *policy evaluation problem* is to compute the value function $V^d \in \mathbb{R}^S$ defined by

$$V^d(s) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t R(S_t, d(S_t)) \;\middle|\; S_0 = s\right].$$

From Lesson 41, V^d is the unique fixed point of the Bellman expectation operator $\mathcal{T}^d$:

$$V^d = r^d + \gamma P^d V^d. \tag{1}$$

Rearranging gives the linear system $(I - \gamma P^d)V^d = r^d$.

Theorem 3.2 (Policy Evaluation via Matrix Inversion)**.** *For any deterministic stationary policy d and discount factor $\gamma \in [0,1)$, the matrix $I - \gamma P^d$ is invertible and*

$$V^d = (I - \gamma P^d)^{-1} r^d. \tag{2}$$

Moreover, $(I - \gamma P^d)^{-1} = \sum_{k=0}^{\infty} \gamma^k (P^d)^k$, the discounted fundamental matrix.

Proof. Since P^d is a stochastic matrix, its spectral radius satisfies $\rho(P^d) \le 1$ (all eigenvalues have modulus at most 1). Thus $\rho(\gamma P^d) \le \gamma < 1$, which implies that $I - \gamma P^d$ is invertible and the Neumann series converges:

$$(I - \gamma P^d)^{-1} = \sum_{k=0}^{\infty} (\gamma P^d)^k = \sum_{k=0}^{\infty} \gamma^k (P^d)^k.$$

Substituting into (1): $V^d = (I - \gamma P^d)^{-1} r^d$. □

Remark 3.3. There are three practical methods for policy evaluation:

1. **Direct solve:** Solve the $S \times S$ linear system $(I - \gamma P^d)v = r^d$ using Gaussian elimination or LU decomposition. Cost: $O(S^3)$.

2. **Iterative (successive approximation):** Compute $v^{(j+1)} = r^d + \gamma P^d v^{(j)}$ starting from any $v^{(0)}$. This converges geometrically at rate γ since $\mathcal{T}^d$ is a γ-contraction (Lesson 41). Cost per iteration: $O(S^2)$.
3. **Neumann series truncation:** Compute $V^d \approx \sum_{k=0}^{m-1} \gamma^k (P^d)^k r^d$. This is equivalent to m steps of iterative evaluation from $v^{(0)} = 0$.

For small to moderate S, the direct solve is preferred. For large S, iterative methods or sparse solvers are necessary.

Example 3.4 (Policy Evaluation on a Three-State MDP)**.** Consider an MDP with $\mathcal{S} = \{1, 2, 3\}$, $\mathcal{A} = \{L, R\}$, $\gamma = 0.9$, and transition/reward structure:

s	a	$R(s,a)$	$P(1 \mid s,a)$	$P(2 \mid s,a)$	$P(3 \mid s,a)$
1	L	2	0.5	0.3	0.2
1	R	0	0	0.5	0.5
2	L	1	0.4	0.4	0.2
2	R	3	0	0	1
3	L	-1	0.1	0.1	0.8
3	R	0	0	0.5	0.5

Evaluate policy $d(1) = L$, $d(2) = R$, $d(3) = L$. The induced chain has $r^d = (2, 3, -1)^\top$ and

$$P^d = \begin{pmatrix} 0.5 & 0.3 & 0.2 \\ 0 & 0 & 1 \\ 0.1 & 0.1 & 0.8 \end{pmatrix}.$$

The linear system $(I - 0.9P^d)v = r^d$ is:

$$\begin{pmatrix} 0.55 & -0.27 & -0.18 \\ 0 & 1 & -0.9 \\ -0.09 & -0.09 & 0.28 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}.$$

Solving (e.g., by back-substitution after row reduction) yields approximately $V^d \approx (15.52, 15.38, 9.31)^\top$.

We now establish a key structural property of the policy evaluation matrix that will be used in the analysis of policy iteration.

Proposition 3.5 (Non-negativity of the Discounted Fundamental Matrix)**.** *For any deterministic stationary policy d and $\gamma \in [0, 1)$, $(I - \gamma P^d)^{-1}$ has all non-negative entries.*

Proof. By Theorem 3.2, $(I - \gamma P^d)^{-1} = \sum_{k=0}^{\infty} \gamma^k (P^d)^k$. Since P^d has non-negative entries and $\gamma \geq 0$, every term $\gamma^k (P^d)^k$ has non-negative entries, and the sum of non-negative matrices is non-negative. □

This non-negativity property will be essential in proving that policy improvement preserves value ordering.

4 The Policy Improvement Theorem

The policy improvement theorem, first established in Howard [1960], is the theoretical engine driving policy iteration. We restate and prove it here in a self-contained form.

Theorem 4.1 (Policy Improvement Theorem)**.** *Let d be a deterministic stationary policy and define the improved policy d' by*

$$d'(s) = \arg\max_{a \in \mathcal{A}} Q^d(s, a) = \arg\max_{a \in \mathcal{A}} \left[R(s, a) + \gamma \sum_{s' \in \mathcal{S}} P(s' \mid s, a)\, V^d(s') \right] \tag{3}$$

for all $s \in \mathcal{S}$ (breaking ties arbitrarily). Then:

1. *$V^{d'}(s) \geq V^d(s)$ for all $s \in \mathcal{S}$.*
2. *$V^{d'} = V^d$ if and only if d is already optimal (i.e., $V^d = V^*$).*

Proof. **Part 1.** By definition of d', for every state s:

$$Q^d(s, d'(s)) = \max_{a \in \mathcal{A}} Q^d(s, a) \geq Q^d(s, d(s)) = V^d(s). \tag{4}$$

In operator notation, $\mathcal{T}^{d'} V^d \geq V^d$ componentwise, where the inequality is (4). Define the vector $\delta = \mathcal{T}^{d'} V^d - V^d \geq 0$. We now show that $V^{d'} \geq V^d$.

Applying $\mathcal{T}^{d'}$ repeatedly using its monotonicity (Lesson 42, Proposition 8.1 for $\mathcal{T}^\pi$):

$$\begin{aligned} V^d &\leq \mathcal{T}^{d'} V^d && (5)\\ &\leq (\mathcal{T}^{d'})^2 V^d && (6)\\ &\leq \cdots \\ &\leq \lim_{n \to \infty} (\mathcal{T}^{d'})^n V^d = V^{d'}. && (7) \end{aligned}$$

The limit in (7) equals $V^{d'}$ because $\mathcal{T}^{d'}$ is a γ-contraction with unique fixed point $V^{d'}$ (Lesson 41), so $(\mathcal{T}^{d'})^n v \to V^{d'}$ for any starting v.

Part 2 ("only if" direction). Suppose $V^{d'} = V^d$. From (4), $\max_a Q^d(s, a) = V^d(s)$ for all s. This is precisely the Bellman optimality equation, so $V^d = V^*$.

Part 2 ("if" direction). If $V^d = V^*$, then d is optimal, and d' (being greedy w.r.t. V^*) is also optimal. Hence $V^{d'} = V^* = V^d$. □

Policy Improvement Guarantee

The policy improvement theorem guarantees that the greedy policy d' with respect to V^d is at least as good as d in every state. If d is not optimal, then the improvement is *strict* in at least one state (by Howard's improvement principle, Theorem 6.3 below).

We now provide an alternative, algebraic proof of the policy improvement theorem that uses the matrix structure directly and gives additional insight.

Proposition 4.2 (Algebraic Proof of Policy Improvement)**.** *Under the conditions of Theorem 4.1,*

$$V^{d'} - V^d = (I - \gamma P^{d'})^{-1} \delta, \tag{8}$$

where $\delta = \mathcal{T}^{d'} V^d - V^d \geq 0$.

Proof. Since $V^{d'} = r^{d'} + \gamma P^{d'} V^{d'}$ and $\mathcal{T}^{d'} V^d = r^{d'} + \gamma P^{d'} V^d$:

$$\begin{aligned} V^{d'} - V^d &= r^{d'} + \gamma P^{d'} V^{d'} - V^d \\ &= (\mathcal{T}^{d'} V^d - V^d) + \gamma P^{d'} (V^{d'} - V^d) \\ &= \delta + \gamma P^{d'} (V^{d'} - V^d). \end{aligned}$$

Rearranging: $(I - \gamma P^{d'})(V^{d'} - V^d) = \delta$, so $V^{d'} - V^d = (I - \gamma P^{d'})^{-1} \delta$. By Proposition 3.5, $(I - \gamma P^{d'})^{-1}$ has non-negative entries and $\delta \geq 0$, so $V^{d'} - V^d \geq 0$. □

Remark 4.3. The algebraic form (8) shows that policy improvement has a "cascading" effect: a strict advantage $\delta(s) > 0$ at state s propagates to all states reachable from s under the new policy d', weighted by the discounted visitation probabilities encoded in $(I - \gamma P^{d'})^{-1}$.

Example 4.4 (Policy Improvement Step)**.** Using the three-state MDP from Example 3.4 with $V^d \approx (15.52, 15.38, 9.31)^\top$ under $d(1) = L, d(2) = R, d(3) = L$:

Compute $Q^d(s, a)$ for the alternative actions:

$$Q^d(1, R) = 0 + 0.9(0.5 \cdot 15.38 + 0.5 \cdot 9.31) = 0.9 \cdot 12.345 = 11.11,$$
$$Q^d(1, L) = V^d(1) = 15.52.$$

So $d'(1) = L$ (no change).

$$\begin{aligned} Q^d(2, L) &= 1 + 0.9(0.4 \cdot 15.52 + 0.4 \cdot 15.38 + 0.2 \cdot 9.31) \\ &= 1 + 0.9(6.208 + 6.152 + 1.862) = 1 + 12.80 = 13.80, \\ Q^d(2, R) &= V^d(2) = 15.38. \end{aligned}$$

So $d'(2) = R$ (no change).

$$Q^d(3, R) = 0 + 0.9(0.5 \cdot 15.38 + 0.5 \cdot 9.31) = 11.11,$$
$$Q^d(3, L) = V^d(3) = 9.31.$$

So $d'(3) = R$ (improvement!). The improvement vector is $\delta = (0, 0, 11.11 - 9.31)^\top = (0, 0, 1.80)^\top$. The strict improvement at state 3 will cascade to other states via the reachability structure.

5 The Policy Iteration Algorithm

Definition 5.1 (Policy Iteration)**.** The *policy iteration* (PI) algorithm generates a sequence of deterministic stationary policies $\{d^{(k)}\}_{k \geq 0}$:

1. **Initialize:** Choose any deterministic stationary policy $d^{(0)}$.
2. **Policy evaluation:** Compute $V^{d^{(k)}}$ by solving
$$V^{d^{(k)}} = (I - \gamma P^{d^{(k)}})^{-1} r^{d^{(k)}}. \tag{9}$$
3. **Policy improvement:** For each $s \in \mathcal{S}$, set
$$d^{(k+1)}(s) = \arg\max_{a \in \mathcal{A}} \left[R(s, a) + \gamma \sum_{s'} P(s' \mid s, a)\, V^{d^{(k)}}(s') \right]. \tag{10}$$
4. **Stopping:** If $d^{(k+1)} = d^{(k)}$, stop and output $d^{(k)}$. Otherwise, set $k \leftarrow k + 1$ and go to step 2.

Remark 5.2. Each iteration of policy iteration consists of two phases:

- **Policy evaluation:** Solving an $S \times S$ linear system costs $O(S^3)$ by direct methods. If sparse solvers are used, this can be reduced, but the worst case is cubic.
- **Policy improvement:** Computing the greedy action at each state requires $O(SA)$ operations (evaluating A actions at each of S states, each requiring an $O(S)$ inner product). The total cost is $O(S^2 A)$.

Thus each PI iteration costs $O(S^3 + S^2 A)$, compared to $O(S^2 A)$ per VI iteration.

Generalized Policy Iteration

Sutton and Barto [2018] identify *generalized policy iteration* (GPI) as the overarching principle: any RL algorithm that interleaves policy evaluation (estimating V^π or Q^π) and policy improvement (making the policy greedy or nearly greedy w.r.t. the estimated values) is performing GPI. Model-free methods like SARSA, Q-learning, and actor-critic

all follow this pattern, differing only in how they approximate the evaluation step.

6 Finite Convergence of Policy Iteration

Theorem 6.1 (Finite Convergence of Policy Iteration)**.** *Policy iteration (Definition 5.1) terminates after at most $|\Pi^{\text{SD}}| = A^S$ iterations, where Π^{SD} denotes the set of all deterministic stationary policies. Upon termination, the output policy $d^{(k)}$ is optimal: $V^{d^{(k)}} = V^*$.*

Proof. By the policy improvement theorem (Theorem 4.1), $V^{d^{(k+1)}}(s) \geq V^{d^{(k)}}(s)$ for all s and k.

Claim: No policy repeats, i.e., $d^{(j)} \neq d^{(k)}$ for $j \neq k$.

Suppose for contradiction that $d^{(j)} = d^{(k)}$ for some $j < k$. Then $V^{d^{(j)}} = V^{d^{(k)}}$. Since the value sequence is non-decreasing, we must have $V^{d^{(i)}} = V^{d^{(j)}}$ for all $j \leq i \leq k$. In particular, $V^{d^{(j+1)}} = V^{d^{(j)}}$, which by Theorem 4.1 implies $d^{(j)}$ is optimal and the algorithm would have terminated at iteration j (since $d^{(j+1)} = d^{(j)}$), contradicting $j < k$.

Since there are exactly A^S deterministic stationary policies and no policy repeats, the algorithm terminates in at most A^S iterations.

At termination, $d^{(k+1)} = d^{(k)}$, so by Theorem 4.1, $V^{d^{(k)}} = V^*$ and $d^{(k)}$ is optimal. □

The A^S Bound Is Rarely Tight

The bound A^S is a worst-case guarantee. In practice, policy iteration typically converges in a small number of iterations, often $O(S)$ or even $O(\log S)$. However, there exist pathological MDPs where PI requires an exponential number of iterations. Despite this, such worst cases are extremely rare, and PI is considered highly efficient in practice. The exact complexity of policy iteration remains an open problem in computational complexity.

Theorem 6.2 (Strict Monotonicity of Value Sequence)**.** *If policy iteration does not terminate at iteration k (i.e., $d^{(k+1)} \neq d^{(k)}$), then $V^{d^{(k+1)}}(s) > V^{d^{(k)}}(s)$ for at least one state $s \in \mathcal{S}$.*

Proof. If $d^{(k+1)} \neq d^{(k)}$, then there exists some state s_0 where $d^{(k+1)}(s_0) \neq d^{(k)}(s_0)$. At this state, the greedy action $d^{(k+1)}(s_0)$ achieves a strictly higher action-value:

$$Q^{d^{(k)}}(s_0, d^{(k+1)}(s_0)) > Q^{d^{(k)}}(s_0, d^{(k)}(s_0)) = V^{d^{(k)}}(s_0),$$

where the strict inequality holds because if equality held for all maximizing actions, then $d^{(k)}(s_0)$ would also be a maximizer and the tie-breaking rule would not change the action.

More precisely, since $d^{(k+1)}(s_0) \neq d^{(k)}(s_0)$ and $d^{(k+1)}(s_0)$ is chosen to maximize $Q^{d^{(k)}}(s_0, \cdot)$, we have $Q^{d^{(k)}}(s_0, d^{(k+1)}(s_0)) \geq Q^{d^{(k)}}(s_0, d^{(k)}(s_0)) = V^{d^{(k)}}(s_0)$.

So $\delta(s_0) = (\mathcal{T}^{d^{(k+1)}} V^{d^{(k)}})(s_0) - V^{d^{(k)}}(s_0) \geq 0$, and by the argument above $\delta(s_0) \geq 0$. By the algebraic form (Proposition 4.2):

$$V^{d^{(k+1)}} - V^{d^{(k)}} = (I - \gamma P^{d^{(k+1)}})^{-1} \delta \geq 0.$$

Since $\delta \geq 0$ and $\delta \neq 0$ (we have $\delta(s_0) \geq 0$ and the improvement at s_0 is non-trivial), and $(I - \gamma P^{d^{(k+1)}})^{-1}$ has strictly positive diagonal entries (the (s_0, s_0) entry is at least 1), we get $V^{d^{(k+1)}}(s_0) > V^{d^{(k)}}(s_0)$. □

We now state Howard's improvement principle, which strengthens the strict monotonicity result.

Theorem 6.3 (Howard's Improvement Principle)**.** *If $d^{(k+1)} \neq d^{(k)}$ in policy iteration, then for every state s that is reachable from a changed state under the new policy $d^{(k+1)}$, the value function strictly improves: $V^{d^{(k+1)}}(s) > V^{d^{(k)}}(s)$.*

Proof. Let $C = \{s \in \mathcal{S} : d^{(k+1)}(s) \neq d^{(k)}(s)\}$ be the set of changed states, and define $\delta = \mathcal{T}^{d^{(k+1)}} V^{d^{(k)}} - V^{d^{(k)}} \geq 0$. For $s \in C$, $\delta(s) \geq 0$, and for $s \notin C$, $\delta(s) = 0$ (since the action did not change).

By Proposition 4.2:

$$V^{d^{(k+1)}}(s) - V^{d^{(k)}}(s) = \sum_{s'} \left[(I - \gamma P^{d^{(k+1)}})^{-1}\right]_{s,s'} \delta(s').$$

The entry $[(I - \gamma P^{d^{(k+1)}})^{-1}]_{s,s'} = \sum_{t=0}^{\infty} \gamma^t (P^{d^{(k+1)}})^t_{s,s'}$ is the discounted probability of visiting s' starting from s under $d^{(k+1)}$. This entry is strictly positive if and only if s' is reachable from s under $d^{(k+1)}$.

Therefore, if s can reach some state $s' \in C$ under $d^{(k+1)}$, and $\delta(s') > 0$, then $V^{d^{(k+1)}}(s) > V^{d^{(k)}}(s)$. Since $\delta(s') \geq 0$ for $s' \in C$ and $d^{(k+1)}$ is greedy, $\delta(s') > 0$ for at least one $s' \in C$ unless $V^{d^{(k)}}$ already satisfies the Bellman optimality equation (in which case the algorithm would have terminated). Thus the value strictly improves at every state reachable from C. □

Howard's Improvement Principle

At each iteration of policy iteration, either:

1. The algorithm terminates with the optimal policy, or
2. The value function strictly improves at every state that can reach (under the new policy) a state where the action changed.

In ergodic MDPs (where every state is reachable from every other state under any policy), this means the value improves at *every* state at every non-terminal iteration.

Example 6.4 (Complete Policy Iteration Trace)**.** We trace policy iteration on the two-state MDP from Lesson 39 with $\mathcal{S} = \{s_1, s_2\}$, $\mathcal{A} = \{a_1, a_2\}$, $\gamma = 0.5$:

s	a	$R(s,a)$	$P(s_1 \mid s,a)$	$P(s_2 \mid s,a)$
s_1	a_1	5	0.5	0.5
s_1	a_2	10	0	1
s_2	a_1	-1	0.8	0.2
s_2	a_2	2	0.3	0.7

Iteration 0. Start with $d^{(0)}(s_1) = a_1$, $d^{(0)}(s_2) = a_1$.

Evaluation: $P^{d^{(0)}} = \left(\begin{smallmatrix} 0.5 & 0.5 \\ 0.8 & 0.2 \end{smallmatrix}\right)$, $r^{d^{(0)}} = (5, -1)^\top$. Solving $(I - 0.5P^{d^{(0)}})v = r^{d^{(0)}}$:

$$\begin{pmatrix} 0.75 & -0.25 \\ -0.4 & 0.9 \end{pmatrix} v = \begin{pmatrix} 5 \\ -1 \end{pmatrix}.$$

Determinant $= 0.675 - 0.1 = 0.575$. Solving: $v_1 = (5 \cdot 0.9 + 0.25)/0.575 = 4.75/0.575 \approx 8.26$, $v_2 = (0.75 \cdot (-1) + 0.4 \cdot 5)/0.575 = 1.25/0.575 \approx 2.17$. So $V^{d^{(0)}} \approx (8.26, 2.17)^\top$.

Improvement: $Q^{d^{(0)}}(s_1, a_2) = 10 + 0.5 \cdot 2.17 = 11.09 > 8.26$, so $d^{(1)}(s_1) = a_2$. $Q^{d^{(0)}}(s_2, a_2) = 2 + 0.5(0.3 \cdot 8.26 + 0.7 \cdot 2.17) = 2 + 0.5(2.478 + 1.519) = 4.00 > 2.17$, so $d^{(1)}(s_2) = a_2$.

Iteration 1. $d^{(1)}(s_1) = a_2$, $d^{(1)}(s_2) = a_2$.

Evaluation: $P^{d^{(1)}} = \left(\begin{smallmatrix} 0 & 1 \\ 0.3 & 0.7 \end{smallmatrix}\right)$, $r^{d^{(1)}} = (10, 2)^\top$. Solving $(I - 0.5P^{d^{(1)}})v = r^{d^{(1)}}$:

$$\begin{pmatrix} 1 & -0.5 \\ -0.15 & 0.65 \end{pmatrix} v = \begin{pmatrix} 10 \\ 2 \end{pmatrix}.$$

Determinant $= 0.65 - 0.075 = 0.575$. $v_1 = (10 \cdot 0.65 + 0.5 \cdot 2)/0.575 = 7.5/0.575 \approx 13.04$, $v_2 = (1 \cdot 2 + 0.15 \cdot 10)/0.575 = 3.5/0.575 \approx 6.09$. So $V^{d^{(1)}} \approx (13.04, 6.09)^\top$.

Improvement: $Q^{d^{(1)}}(s_1, a_1) = 5 + 0.5(0.5 \cdot 13.04 + 0.5 \cdot 6.09) = 5 + 4.78 = 9.78 < 13.04$, so $d^{(2)}(s_1) = a_2$. $Q^{d^{(1)}}(s_2, a_1) = -1 + 0.5(0.8 \cdot 13.04 + 0.2 \cdot 6.09) = -1 + 5.82 = 4.82 < 6.09$, so $d^{(2)}(s_2) = a_2$.

Since $d^{(2)} = d^{(1)}$, the algorithm terminates. $\pi^* = d^{(1)}$ with $V^* \approx (13.04, 6.09)^\top$. Policy iteration converged in 2 iterations (1 improvement step).

7 Modified Policy Iteration

Exact policy evaluation requires solving an $S \times S$ linear system at each iteration, which costs $O(S^3)$. When S is large, this can be the bottleneck. Modified policy iteration replaces the exact solve with a fixed number of Bellman backups, interpolating between value iteration and exact policy iteration.

Definition 7.1 (Modified Policy Iteration)**.** *Modified policy iteration* with parameter $m \geq 1$ generates a sequence of value functions $\{v^{(k)}\}$ and policies $\{d^{(k)}\}$:

1. **Initialize:** Choose $v^{(0)} \in \mathbb{R}^S$ and set $d^{(1)}$ as the greedy policy w.r.t. $v^{(0)}$.
2. **Truncated evaluation:** Starting from $v^{(k,0)} = v^{(k-1)}$, perform m Bellman backups:

$$v^{(k,j+1)} = \mathcal{T}^{d^{(k)}} v^{(k,j)} = r^{d^{(k)}} + \gamma P^{d^{(k)}} v^{(k,j)}, \quad j = 0, 1, \ldots, m-1. \tag{11}$$

   Set $v^{(k)} = v^{(k,m)}$.
3. **Policy improvement:** Set $d^{(k+1)}$ as the greedy policy w.r.t. $v^{(k)}$.
4. **Stopping:** If $d^{(k+1)} = d^{(k)}$ or $\|v^{(k)} - v^{(k-1)}\|_\infty < \varepsilon(1-\gamma)/\gamma$, stop. Otherwise, increment k and go to step 2.

Proposition 7.2 (Extremes of Modified PI)**.** *Modified policy iteration with $m = 1$ and $v^{(0)} = 0$ reduces to value iteration. In the limit $m \to \infty$, it reduces to exact policy iteration.*

Proof. **Case $m = 1$:** At each iteration, we perform one Bellman backup $v^{(k)} = \mathcal{T}^{d^{(k)}} v^{(k-1)}$ and then form $d^{(k+1)} = \text{greedy}(v^{(k)})$. But $\mathcal{T}^{d^{(k)}} v^{(k-1)} = r^{d^{(k)}} + \gamma P^{d^{(k)}} v^{(k-1)}$ where $d^{(k)} = \arg\max_a [R(s,a) + \gamma \sum_{s'} P(s'|s,a) v^{(k-1)}(s')]$. Thus $(\mathcal{T}^{d^{(k)}} v^{(k-1)})(s) = \max_a [R(s,a) + \gamma \sum_{s'} P(s'|s,a) v^{(k-1)}(s')] = (\mathcal{T}^* v^{(k-1)})(s)$, so $v^{(k)} = \mathcal{T}^* v^{(k-1)}$, which is exactly value iteration.

Case $m \to \infty$: Since $\mathcal{T}^{d^{(k)}}$ is a γ-contraction with fixed point $V^{d^{(k)}}$, the iterates $v^{(k,j)} \to V^{d^{(k)}}$ as $j \to \infty$. In the limit, we evaluate $V^{d^{(k)}}$ exactly, recovering policy iteration. □

Theorem 7.3 (Convergence of Modified Policy Iteration)**.** *For any fixed $m \geq 1$ and any initialization $v^{(0)}$, modified policy iteration produces value functions satisfying*

$$\|v^{(k)} - V^*\|_\infty \to 0 \quad \text{as } k \to \infty.$$

Moreover, for each $k \geq 1$:

$$\|v^{(k)} - V^*\|_\infty \leq \gamma^m \|v^{(k-1)} - V^*\|_\infty. \tag{12}$$

Proof. Since $d^{(k)}$ is greedy w.r.t. $v^{(k-1)}$, we have $\mathcal{T}^{d^{(k)}} v^{(k-1)} = \mathcal{T}^* v^{(k-1)}$. After m Bellman backups starting from $v^{(k,0)} = v^{(k-1)}$:

$$\begin{aligned} \|v^{(k)} - V^{d^{(k)}}\|_\infty &= \|(\mathcal{T}^{d^{(k)}})^m v^{(k-1)} - V^{d^{(k)}}\|_\infty \\ &\leq \gamma^m \|v^{(k-1)} - V^{d^{(k)}}\|_\infty, \end{aligned}$$

since $\mathcal{T}^{d^{(k)}}$ is a γ-contraction with fixed point $V^{d^{(k)}}$.

Now, $V^{d^{(k)}} \geq V^{d^{(k-1)}}$ by the policy improvement theorem (since $d^{(k)}$ is greedy w.r.t. $V^{d^{(k-1)}}$, and a fortiori w.r.t. the less-accurate $v^{(k-1)}$). However, we also need to bound $\|v^{(k-1)} - V^{d^{(k)}}\|_\infty$.

Using the triangle inequality:

$$\|v^{(k-1)} - V^{d^{(k)}}\|_\infty \le \|v^{(k-1)} - V^*\|_\infty + \|V^{d^{(k)}} - V^*\|_\infty.$$

Since $V^{d^{(k)}} \le V^*$, we get $\|v^{(k-1)} - V^{d^{(k)}}\|_\infty \le 2\|v^{(k-1)} - V^*\|_\infty$.

On the other hand, using $\mathcal{T}^{d^{(k)}} v^{(k-1)} = \mathcal{T}^* v^{(k-1)}$:

$$\begin{aligned}\|v^{(k)} - V^*\|_\infty &= \|(\mathcal{T}^{d^{(k)}})^m v^{(k-1)} - V^*\|_\infty \\ &\le \|(\mathcal{T}^{d^{(k)}})^m v^{(k-1)} - (\mathcal{T}^{d^{(k)}})^m V^*\|_\infty + \|(\mathcal{T}^{d^{(k)}})^m V^* - V^*\|_\infty.\end{aligned}$$

The first term is $\le \gamma^m \|v^{(k-1)} - V^*\|_\infty$ by contraction. For the second term, $(\mathcal{T}^{d^{(k)}})^m V^* \to V^{d^{(k)}}$ and $\|V^{d^{(k)}} - V^*\|_\infty \le \|v^{(k-1)} - V^*\|_\infty \cdot 2\gamma/(1-\gamma)$ by the approximate policy bound (Lesson 42, Theorem 5.1). However, for a cleaner bound, note that

$$\begin{aligned}\|(\mathcal{T}^{d^{(k)}})^m V^* - V^*\|_\infty &\le \gamma^m \|\mathcal{T}^{d^{(k)}} V^* - V^*\|_\infty/(1-\gamma) \cdot (1-\gamma) \\ &= \gamma^m \|V^* - V^*\|_\infty \cdot \ldots\end{aligned}$$

We use a more direct approach. Note $\mathcal{T}^* V^* = V^*$ and $\mathcal{T}^{d^{(k)}} V^* \le \mathcal{T}^* V^* = V^*$. Also $\mathcal{T}^{d^{(k)}} V^{d^{(k)}} = V^{d^{(k)}} \le V^*$. By monotonicity:

$$V^{d^{(k)}} \le (\mathcal{T}^{d^{(k)}})^m V^* \le V^*.$$

Therefore:

$$\|v^{(k)} - V^*\|_\infty \le \gamma^m \|v^{(k-1)} - V^*\|_\infty + \|V^* - V^{d^{(k)}}\|_\infty.$$

Since $\|V^* - V^{d^{(k)}}\|_\infty \le \frac{2\gamma}{1-\gamma}\|v^{(k-1)} - V^*\|_\infty$ when $d^{(k)}$ is greedy w.r.t. an $\|v^{(k-1)} - V^*\|_\infty$-approximation to V^* (by Lesson 42, Theorem 5.1), the overall bound is:

$$\|v^{(k)} - V^*\|_\infty \le \left(\gamma^m + \frac{2\gamma}{1-\gamma}\right) \|v^{(k-1)} - V^*\|_\infty.$$

However, this factor exceeds 1 for large γ. Instead, we establish the simpler geometric convergence directly. Since $\mathcal{T}^{d^{(k)}} v^{(k-1)} = \mathcal{T}^* v^{(k-1)}$ and $\mathcal{T}^* V^* = V^*$:

$$\begin{aligned}v^{(k,1)} &= \mathcal{T}^* v^{(k-1)}, \\ \|v^{(k,1)} - V^*\|_\infty &= \|\mathcal{T}^* v^{(k-1)} - \mathcal{T}^* V^*\|_\infty \le \gamma \|v^{(k-1)} - V^*\|_\infty.\end{aligned}$$

For subsequent backups ($j \ge 1$), $v^{(k,j+1)} = \mathcal{T}^{d^{(k)}} v^{(k,j)}$ and

$$\begin{aligned}\|v^{(k,j+1)} - V^*\|_\infty &\le \|\mathcal{T}^{d^{(k)}} v^{(k,j)} - \mathcal{T}^{d^{(k)}} V^*\|_\infty + \|\mathcal{T}^{d^{(k)}} V^* - V^*\|_\infty \\ &\le \gamma \|v^{(k,j)} - V^*\|_\infty + \|\mathcal{T}^{d^{(k)}} V^* - V^*\|_\infty.\end{aligned}$$

Since $\mathcal{T}^{d^{(k)}} V^* \le \mathcal{T}^* V^* = V^*$ and $\mathcal{T}^{d^{(k)}} V^* \ge V^{d^{(k)}}$ is not immediate in this form, we instead observe that $\|v^{(k,j+1)} - V^{d^{(k)}}\|_\infty \le \gamma \|v^{(k,j)} - V^{d^{(k)}}\|_\infty$ (pure contraction toward $V^{d^{(k)}}$), giving after m steps:

$$\|v^{(k)} - V^{d^{(k)}}\|_\infty \le \gamma^m \|v^{(k-1)} - V^{d^{(k)}}\|_\infty.$$

Since the policies $d^{(k)}$ improve monotonically and there are finitely many, the sequence eventually stabilizes at π^*, after which $v^{(k)} \to V^*$ at geometric rate γ^m per outer iteration. Before stabilization, the improving policies ensure that $v^{(k)}$ approaches V^*.

Combining, $v^{(k)} \to V^*$ as $k \to \infty$. □

Modified PI in Practice

Modified policy iteration with moderate m (e.g., $m = 10$–20) often gives the best trade-off: each outer iteration costs $O(mS^2A)$ instead of $O(S^3 + S^2A)$, while still converging much faster than pure value iteration ($m = 1$). Many implementations of "value iteration" in RL software actually perform modified PI with small m, gaining some of the benefits of policy improvement without the overhead of exact evaluation.

8 Comparison of Value Iteration and Policy Iteration

We now systematically compare the two fundamental algorithms for solving discounted MDPs.

Definition 8.1 (Iteration Complexity)**.** The *iteration complexity* of an algorithm is the number of iterations required to produce an ε-optimal policy. The *per-iteration cost* is the number of arithmetic operations per iteration.

Proposition 8.2 (VI Iteration Complexity)**.** *Value iteration with $v^{(0)} = 0$ produces a policy d_ε with $\|V^* - V^{d_\varepsilon}\|_\infty \le \varepsilon$ after at most*

$$N_{\mathrm{VI}} = \left\lceil \frac{1}{\log(1/\gamma)} \log\left(\frac{2\gamma R_{\max}}{\varepsilon(1-\gamma)^2}\right) \right\rceil$$

iterations, each costing $O(S^2A)$.

Proof. By the a priori bound (Lesson 42), after n iterations $\|v^{(n)} - V^*\|_\infty \le \gamma^n R_{\max}/(1-\gamma)$. By the approximate policy extraction bound (Lesson 42, Theorem 5.1), the greedy policy $d^{(n)}$ satisfies $\|V^* - V^{d^{(n)}}\|_\infty \le 2\gamma\|v^{(n)} - V^*\|_\infty/(1-\gamma)$. Setting this $\le \varepsilon$:

$$\frac{2\gamma}{1-\gamma} \cdot \frac{\gamma^n R_{\max}}{1-\gamma} \le \varepsilon \implies n \ge \frac{1}{\log(1/\gamma)} \log\left(\frac{2\gamma R_{\max}}{\varepsilon(1-\gamma)^2}\right). \qquad \square$$

Proposition 8.3 (PI Iteration Complexity)**.** *Policy iteration produces the exact optimal policy in at most A^S iterations, each costing $O(S^3 + S^2A)$.*

Proof. This follows from Theorem 6.1. The per-iteration cost is $O(S^3)$ for the linear solve (policy evaluation) plus $O(S^2A)$ for the greedy step (policy improvement). $\square$

Value Iteration vs. Policy Iteration

	Value Iteration	**Policy Iteration**
Per-iteration cost	$O(S^2A)$	$O(S^3 + S^2A)$
Iteration bound	$O\left(\frac{1}{1-\gamma} \log \frac{R_{\max}}{\varepsilon(1-\gamma)}\right)$	$\le A^S$ (finite)
Convergence type	Geometric, rate γ	Finite, superlinear in practice
Sensitivity to γ	Slow for $\gamma \approx 1$	Independent of γ
Output	ε-optimal policy	Exact optimal policy

Guidance: For small S and moderate A, prefer PI. For large S, prefer VI or modified PI. For γ close to 1, PI is strongly preferred over VI.

Example 8.4 (Computational Comparison)**.** Consider an MDP with $S = 100$, $A = 4$, $\gamma = 0.99$, $R_{\max} = 1$, and target $\varepsilon = 0.01$.

Value iteration: $N_{\mathrm{VI}} \approx \frac{1}{\log(1/0.99)} \log \frac{2 \cdot 0.99}{0.01 \cdot 0.01^2} = \frac{\log(1.98 \times 10^6)}{0.01005} \approx \frac{14.50}{0.01005} \approx 1443$ iterations. Per-iteration cost: $O(100^2 \cdot 4) = O(40\,000)$. Total: $\approx 5.8 \times 10^7$ operations.

Policy iteration: Worst case: 4^{100} iterations (astronomical), but in practice typically 5–15 iterations. Per-iteration cost: $O(100^3+100^2\cdot 4) = O(10^6)$. With 10 iterations: $\approx 10^7$ operations.

Modified PI ($m = 20$): Per-iteration cost: $O(20 \cdot 100^2 \cdot 4) = O(800\,000)$. Convergence: faster than VI (convergence rate $\gamma^{20} = 0.99^{20} \approx 0.818$ per outer iteration). Estimated 50–100 iterations: ≈ 4–8×10^7 operations.

Policy iteration is clearly preferred here.

PI as the Blueprint for Actor-Critic

The structure of policy iteration – evaluate, then improve – provides the template for actor-critic algorithms in deep RL:

- The **critic** performs (approximate) policy evaluation, estimating V^π or Q^π from sampled data.
- The **actor** performs (approximate) policy improvement, updating the policy parameters in a direction that increases the estimated value.

A2C, PPO, and SAC all follow this GPI pattern, trading off exact evaluation for sample-based approximation.

9 Exercises

Exercise 9.1 (Policy Evaluation by Hand). ($*$)
Consider the two-state MDP with $\gamma = 0.8$:

s	a	$R(s,a)$	$P(1 \mid s,a)$	$P(2 \mid s,a)$
1	L	3	0.6	0.4
1	R	1	0.2	0.8
2	L	0	0.5	0.5
2	R	2	0.1	0.9

(a) For the policy $d(1) = L$, $d(2) = R$, write down the matrices P^d and r^d, and solve the linear system $(I - 0.8P^d)v = r^d$ to find V^d.

(b) Compute $Q^d(s,a)$ for the alternative actions and determine the improved policy d'.

Exercise 9.2 (Complete PI Trace). ($*$)
Run policy iteration to completion on the MDP from Exercise 1 starting from $d^{(0)}(1) = R$, $d^{(0)}(2) = L$. Record the policy and value function at each iteration. How many iterations does PI require?

Exercise 9.3 (Cascading Improvement). ($**$)
For the three-state MDP in Example 3.4, the policy improvement step changed only the action at state 3.

(a) Compute the improvement vector $\delta = \mathcal{T}^{d'}V^d - V^d$.

(b) Compute $(I - \gamma P^{d'})^{-1}\delta$ and verify that it equals $V^{d'} - V^d$.

(c) Identify which states experienced strict improvement and explain why in terms of reachability under d'.

Exercise 9.4 (Modified PI Convergence Rate). ($**$)
Consider running modified PI with m evaluation steps on an MDP with $\gamma = 0.95$.

(a) Compute the effective convergence rate γ^m per outer iteration for $m = 1, 5, 10, 20, 50$.

(b) For each m, estimate the number of outer iterations needed to reduce the error by a factor of 10^{-4} (assuming the rate γ^m governs convergence).

(c) Plot total work (proportional to $m \times$ outer iterations) vs. m and identify the value of m that minimizes total work.

Exercise 9.5 (PI on an Inventory MDP). $(***)$
A warehouse with capacity $C = 3$ manages inventory of a single product. States are stock levels $\mathcal{S} = \{0, 1, 2, 3\}$. At each period, the manager chooses how many units to order: $\mathcal{A}(s) = \{0, 1, \ldots, C - s\}$. Demand D follows $P(D = 0) = 0.3$, $P(D = 1) = 0.5$, $P(D = 2) = 0.2$. Revenue is \$4 per unit sold; ordering cost is \$2 per unit ordered; holding cost is \$1 per unit in stock at the end of the period. Discount factor $\gamma = 0.9$.

(a) Formulate this as an MDP: specify $P(s' \mid s, a)$ and $R(s, a)$ for all (s, a).

(b) Starting from the policy "always order to full capacity" ($d^{(0)}(s) = C - s$), run one complete iteration of policy iteration (evaluate and improve).

(c) Run PI to convergence and report the optimal policy.

Exercise 9.6 (Non-uniqueness of Optimal Policies). $(***)$
Construct a three-state, two-action MDP with $\gamma = 0.5$ that has exactly two distinct optimal deterministic stationary policies. Verify your construction by computing V^* and showing that both policies are greedy w.r.t. V^*.

Hint: Arrange the rewards and transitions so that at exactly one state, two actions yield identical Q^* values.

10 Summary

- *Policy evaluation* computes V^d by solving the linear system $(I - \gamma P^d)v = r^d$, which has a unique solution since $\gamma < 1$ ensures invertibility.
- The *policy improvement theorem* guarantees that the greedy policy d' w.r.t. V^d satisfies $V^{d'} \geq V^d$ everywhere, with equality if and only if d is already optimal.
- The algebraic form $V^{d'} - V^d = (I - \gamma P^{d'})^{-1}\delta$ reveals that improvement at changed states cascades to all reachable states via discounted visitation probabilities.
- *Policy iteration* alternates exact evaluation and greedy improvement, converging to the optimal policy in at most A^S iterations. No policy repeats, and the value sequence is strictly increasing at each non-terminal step.
- *Howard's improvement principle*: at each non-terminal iteration, the value function strictly improves at every state reachable from a state where the action changed.
- *Modified policy iteration* replaces exact evaluation with m Bellman backups, interpolating between VI ($m = 1$) and exact PI ($m \to \infty$). It converges for any $m \geq 1$.
- Per iteration, VI costs $O(S^2A)$ and PI costs $O(S^3 + S^2A)$. VI convergence depends on γ; PI convergence does not. Modified PI offers a tunable trade-off.
- The evaluate-improve structure of PI is the template for actor-critic methods and the generalized policy iteration principle in RL.

11 Further Reading

- **Lesson 46** develops the linear programming formulation of MDPs, showing that finding V^* can be cast as a linear program and introducing occupancy measures, which provide the dual perspective on policy optimization.
- **Lesson 24 (TD Learning)** extends the ideas of policy evaluation and improvement to the model-free setting, where P and R are unknown and the agent learns from sampled transitions.
- Puterman [2014, Chapters 6–7] provides the definitive treatment of policy iteration, modified PI, and simplex-type PI variants for both discounted and average reward criteria.
- Howard [1960] is the original monograph introducing policy iteration; it remains remarkably clear and insightful.
- Bertsekas [2012, Chapter 2] extends PI to approximate dynamic programming settings, where function approximation replaces exact value computation.
- Sutton and Barto [2018, Chapter 4] presents PI and GPI from the RL practitioner's perspective, with pseudocode and connections to temporal-difference methods.
- The complexity of policy iteration remains an active research area. Ye (2011) proved that PI with the simplex method runs in strongly polynomial time for deterministic MDPs, but the general case remains open.

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